

Reason about arithmetic relationships

Use the scaffold, show each step and ask for an adult check when marked.

Name: _____ Date: _____

1. Read the model: If $4,725 / 35 = 135$, find $4,725 / 350$. Copy or complete the first step.

2. If $48 \times 125 = 6,000$, find 4.8×125 .

3. Which is greater: $5/6$ of 420 or 0.82 of 420?

4. Miro says: Miro changes one factor by ten but expects the product to stay unchanged. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: If $4,725 / 35 = 135$, find $4,725 / 350$. Copy or complete the first step.

Answer 13.5.

2. If $48 \times 125 = 6,000$, find 4.8×125 .

Answer 600.

3. Which is greater: $5/6$ of 420 or 0.82 of 420?

Answer $5/6$ of 420 = 350; 0.82 of 420 = 344.4, so $5/6$ is greater.

4. Miro says: Miro changes one factor by ten but expects the product to stay unchanged. Is this correct? Explain.

Answer Changing a factor by a power of ten changes the product by the same scale unless another factor compensates.

Reason about arithmetic relationships

Work independently. Show a complete method and check at least one answer.

Name: _____ Date: _____

1. If $48 \times 125 = 6,000$, find 4.8×125 .

2. Which is greater: $\frac{5}{6}$ of 420 or 0.82 of 420?

3. Create a chain of three related facts from $2,304 \times 27 = 62,208$.

4. Identify and correct this misconception: Miro changes one factor by ten but expects the product to stay unchanged.

5. Write a complete sentence explaining how you checked one answer.

Teacher answers and checking prompts

1. If $48 \times 125 = 6,000$, find 4.8×125 .

Answer 600.

2. Which is greater: $\frac{5}{6}$ of 420 or 0.82 of 420?

Answer $\frac{5}{6}$ of 420 = 350; 0.82 of 420 = 344.4, so $\frac{5}{6}$ is greater.

3. Create a chain of three related facts from $2,304 \times 27 = 62,208$.

Answer Answers vary, for example $62,208 \div 27 = 2,304$ and $230.4 \times 27 = 6,220.8$.

4. Identify and correct this misconception: Miro changes one factor by ten but expects the product to stay unchanged.

Answer Changing a factor by a power of ten changes the product by the same scale unless another factor compensates.

5. Write a complete sentence explaining how you checked one answer.

Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.

Reason about arithmetic relationships

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: _____ Date: _____

1. Create a chain of three related facts from $2,304 \times 27 = 62,208$.

2. Explain why this reasoning fails: Miro changes one factor by ten but expects the product to stay unchanged.

3. Create a new example that tests the same idea as: Which is greater: $\frac{5}{6}$ of 420 or 0.82 of 420?

4. Find a second method or representation. Compare its efficiency with your first method.

5. Write one always, sometimes or never statement about today's learning and justify it.

Teacher answers and checking prompts

1. Create a chain of three related facts from $2,304 \times 27 = 62,208$.
Answer Answers vary, for example $62,208 \div 27 = 2,304$ and $230.4 \times 27 = 6,220.8$.
2. Explain why this reasoning fails: Miro changes one factor by ten but expects the product to stay unchanged.
Answer Changing a factor by a power of ten changes the product by the same scale unless another factor compensates.
3. Create a new example that tests the same idea as: Which is greater: $\frac{5}{6}$ of 420 or 0.82 of 420?
Answer Answers vary. The example and solution must preserve the mathematical structure.
4. Find a second method or representation. Compare its efficiency with your first method.
Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.
5. Write one always, sometimes or never statement about today's learning and justify it.
Answer Answers vary. A valid example and counterexample should support the classification.